Multivariate Time Series (MTS) Forecasting

The Core Challenge

MTS forecasting requires modelling two distinct but coupled types of dependencies simultaneously:

- **Intra-series (Temporal):** Dependencies within a single time series over time.
- **Inter-series (Spatial):** Correlations between different variables (e.g., traffic sensors, weather stations) at specific timestamps.

The Problem

How to unify these dependencies?
Real-world data exhibits unified spatiotemporal dynamics (e.g., traffic congestion propagation). Treating spatial and temporal modelling as separate steps violates these inherent connections and hinders performance.

Existing Solutions and Why They Lack

Common Approaches:

- **RNNs/CNNs (e.g., LSTM, TCN):** Focus heavily on temporal dependencies; struggle with complex spatial correlations among variables.
- **GNN-based Methods (e.g., STGCN, MTGNN):** Currently the State-of-the-Art. They stack:
    - Graph Networks (spatial)
    - Temporal Networks (temporal)

Key Limitations:

1. **Decoupled Modeling:** Separate spatial/temporal modules create "uncertain compatibility" and high parameter complexity.
2. **High Complexity:** Many GNNs have $O(N^2)$ complexity, making them expensive for large datasets.
3. **Static Priors:** Reliability often depends on pre-defined graph structures (which are often unknown or time-varying).

Novel Solution: The Pure Graph Perspective

Rethinking the Data Structure

Instead of separating dimensions, FourierGNN introduces the **Hypervariate Graph**.

- **Concept:** Treat every single data point $x_t^{(n)}$ (variate n at time t) as a unique node in one massive graph.
- **Structure:** A sliding window of N series over length T becomes a graph with $N \times T$ nodes.
- **Connectivity:** Initialised as a fully-connected graph to capture all possible correlations.

"This perspective considers spatiotemporal dynamics unitedly and reformulates forecasting into predictions on hypervariate graphs."

Novel Solution: Fourier Graph Neural Network (FourierGNN)

To handle the large size of Hypervariate Graphs ($N \times T$ nodes), the authors propose the **Fourier Graph Operator (FGO)**.

1. **Fourier Domain Operation:** Instead of expensive spatial convolution, the model operates in the frequency domain using DFT (Discrete Fourier Transform).

$$\text{Convolution Theorem: } \mathcal{F}(X * \kappa) = \mathcal{F}(X) \odot \mathcal{F}(\kappa)$$

2. **Efficiency:**
   - Spatial Graph Convolution: $O(N^2)$ complexity.
   - **FourierGNN:** Reduces complexity to $O(N \log N)$ (Log-linear) via FFT.
3. **Mechanism:** Stacking FGO layers in the Fourier space is mathematically equivalent to performing multi-order graph convolutions in the time domain, capturing long-range dependencies efficiently.

Key Results and Performance

Accuracy:

- Evaluated on 7 benchmarks (Traffic, Solar, Electricity, etc.).
- Achieved State-of-the-Art (SOTA) performance, outperforming Transformers (Informer) and GNNs (GraphWaveNet).
- **10% average accuracy improvement** reported in the paper.

Efficiency:

- **Fewer Parameters:** $\approx 20\%$ fewer parameters than lightweight baselines.
- **Faster Training:** Significant reduction in training time due to $O(N \log N)$ complexity.

Reproduction Analysis

Qualitative analysis confirms the model's ability to capture periodic patterns, though performance is sensitive to data preprocessing (scaling/normalisation).

Future Directions

Limitations of FourierGNN

- **Loss of Sequential Information:** Treating every time-step as a node makes the graph *order-insensitive*. Permuting the time sequence results in an identical graph structure, losing temporal context.
- **Attention Dilution:** The **Fully-Connected** structure treats all connections uniformly. This "over-smooths" the data, washing out distinct local patterns with global noise.

What we think?

Proposal: Replace Hypervariate Graphs with **Hyperspectral Graphs**.

- **The Method:** Embeds sequential order into frequency signals *before* graph construction.
- **Intuitive Benefit:**
  - *Preserves Time:* Embedding frequency locks the sequential order.
  - *Sharper Focus:* Allow the model to capture the global trends without blurring out specific, local features.

Contributions

Ayush Rawat - Report, Dataset Sourcing, Code
Omkar Bhandare - Code, Future Directions
Shivam Choudhury - Dataset Sourcing, Future Directions, Code